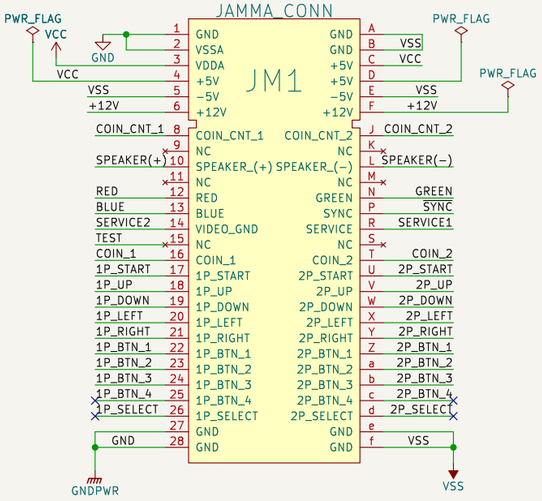
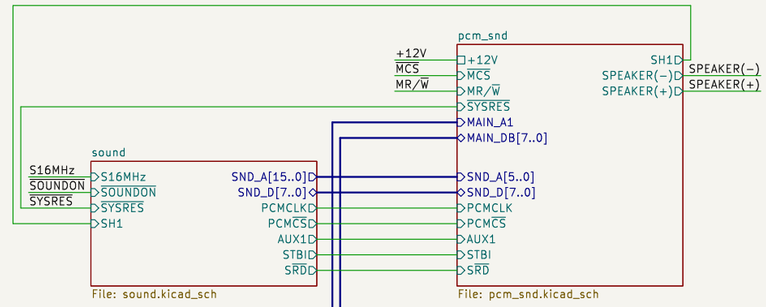
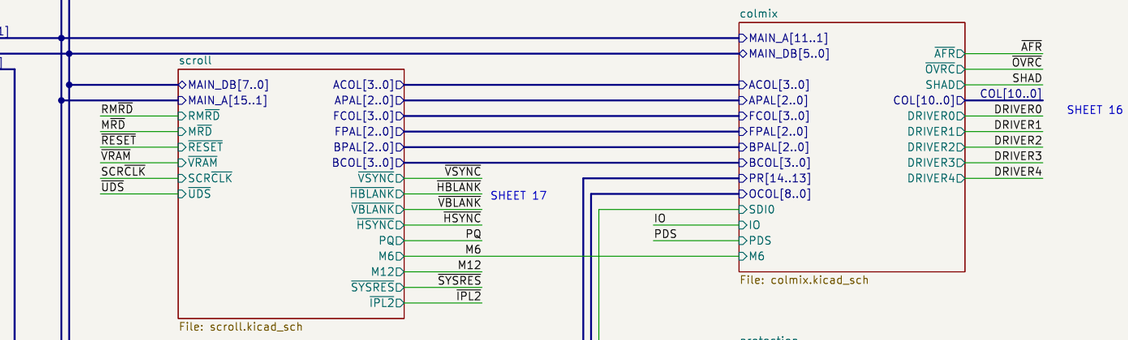
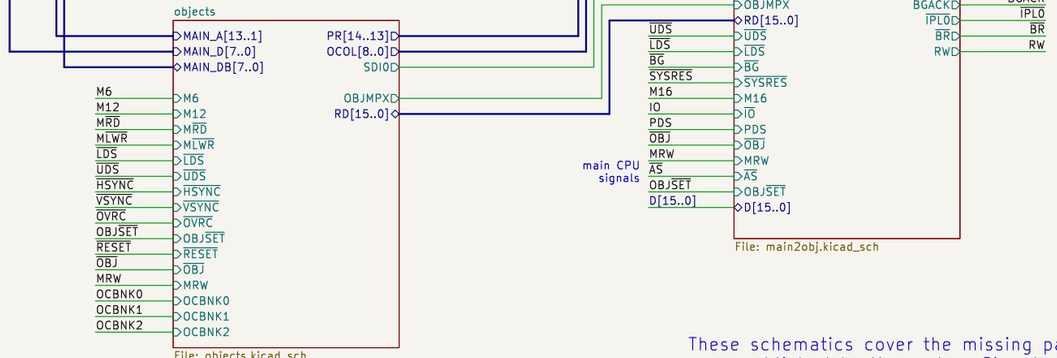








PDB[7:0] is either main CPU upper or lower byte depending on UDS/LDS See sch. page 18, D8/E8

PDS=UDS&LDS

These schematics cover the missing pages in the original ones published by the maker. Signal names existing in the original pages have been preserved. New signal names try to follow the maker style or are taken from similar schematics of the same maker.

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

Sheet: /
File: ssriders.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 1/7

MAIN_A[11..1]

G13

SHEET 16

MAIN_A8 1
MAIN_A9 2
MAIN_A10 3
PDSD 6
IOD 5
MAIN_A11 4

A0 A1 A2 address decoder E3 E2 E1

O0 O1 O2 O3 O4 O5 O6 O7

15 DRIVER0 PIN1 B13
14 DRIVER1 PIN15 C18
13 DRIVER2 PIN11 C15
12 DRIVER3 PIN9 C14
11 AFR
10 OVRC
9 DRIVER4 PIN5 F18
7 PCUCS

74LS138

On the schematic it appears as 17F,
but on the PCB it is 18F.

MAIN_DB[5..0]

PR[14..13]

VCC

PR14 PR13

VSS

MAIN_DB0 MAIN_DB1 MAIN_DB2 MAIN_DB3 MAIN_DB4 MAIN_DB5 MAIN_A1 MAIN_A2 MAIN_A3 MAIN_A4 CSDIO

6M M6

VSS VSS

27 25 24 23 22 2 1 100 99 98 97 80 79 77 76 75 74 85 84 83 82 94 95 96 41

PR15 PR14 PR13 PR12 PR11 PR10 PR05 PR04 PR03 PR02 PR01 PR00 D0 D1 D2 D3 D4 D5 A0 A1 A2 A3 SEL SDIO SDI1 CLK

29 PR15
39 PR20
42 PR21
43 PR22
44 PR23
45 PR24
46 PR25

PRIORITY

VSS 4
5 CI00
6 CI01
7 CI02
8 CI03
9 CI04
10 CI05
20 CI06
21 CI07
30 CI08
31 CI10
32 CI11
33 CI12
34 CI13
35 CI14
36 CI15
37 CI16
38 CI17
CI18

COLOR INPUTS

OCOL0
OCOL1
OCOL2
OCOL3
OCOL4
OCOL5
OCOL6
OCOL7
OCOL8

OCOL[8..0]

J4

KONAMI

053251

PRIORITY CONTROL UNIT

30 pins top/bottom
20 pins left/right

3 VCC
78 VCC
53 VCC
28 VCC
26 N.C

CHIP SELECT INPUT
COLOR OUTPUTS

81 CS
19 CO00
18 CO01
17 CO02
16 CO03
14 CO04
13 CO05
12 CO06
11 CO07
93 CO08
92 CO09
91 CO10
86 SDO1
87 SDO0
88 BRIT
89 NCOL
15 VSS
40 VSS
65 VSS
90 VSS

PCUCS
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
COL9
COL10

COL[10..0]

SHAD

BRIT pdf page 17

VSS

CI20 CI21 CI22 CI23 CI24 CI25 CI26 CI27 CI28 CI30 CI31 CI32 CI33 CI34 CI35 CI36 CI37 CI40 CI41 CI42 CI43 CI44 CI45 CI46 CI47

47 48 49 50 51 52 54 55 56 57 58 59 60 61 62 63 64 66 67 68 69 70 71 72 73

FCOL0 FCOL1 FCOL2 FCOL3 FPAL0 FPAL1 FPAL2

FCOL[3..0]
FPAL[2..0]

ACOL0 ACOL1 ACOL2 ACOL3 APAL0 APAL1 APAL2

ACOL[3..0]
APAL[2..0]

BCOL0 BCOL1 BCOL2 BCOL3 BPAL0 BPAL1 BPAL2

BCOL[3..0]
BPAL[2..0]

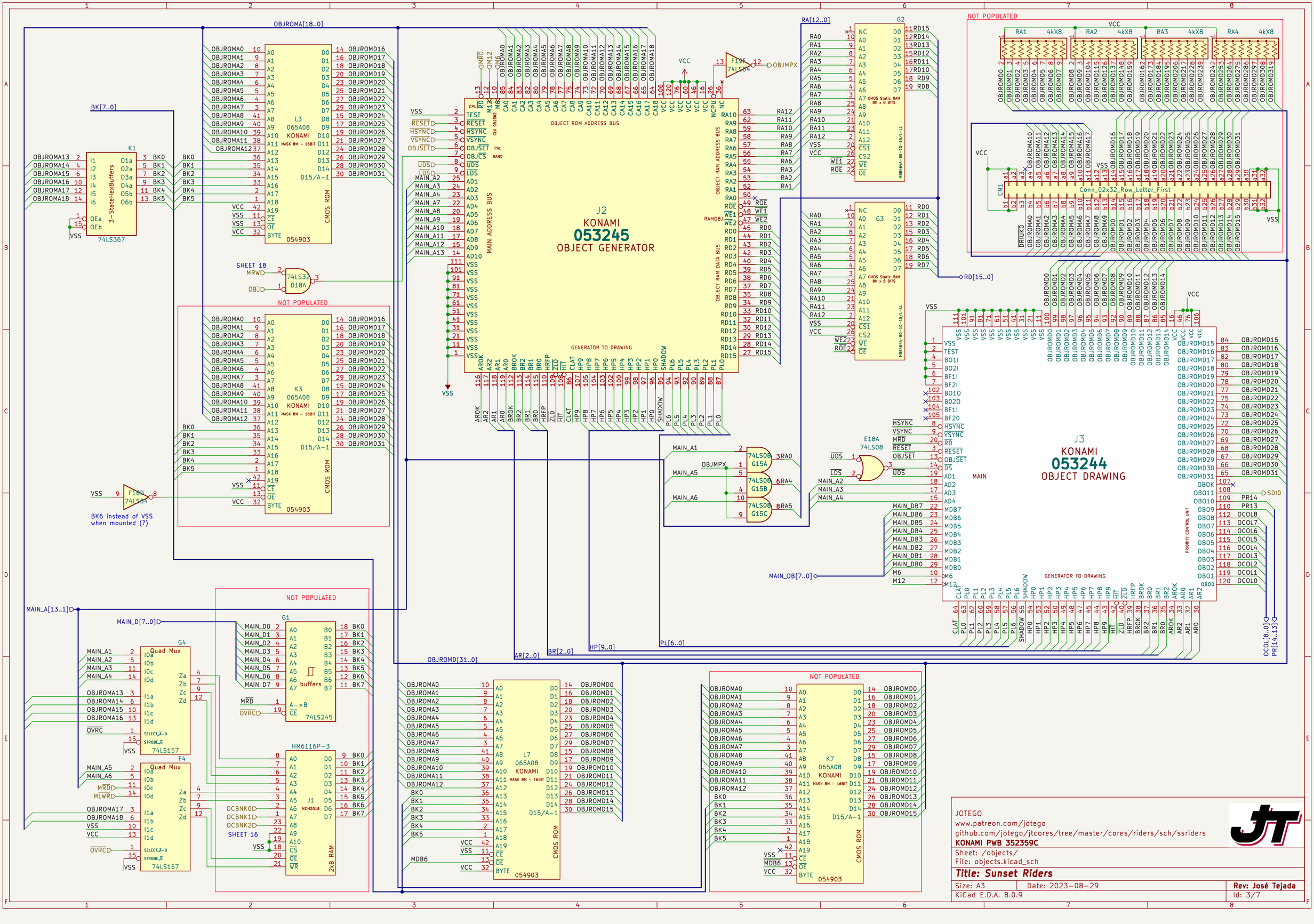
SHEET 02

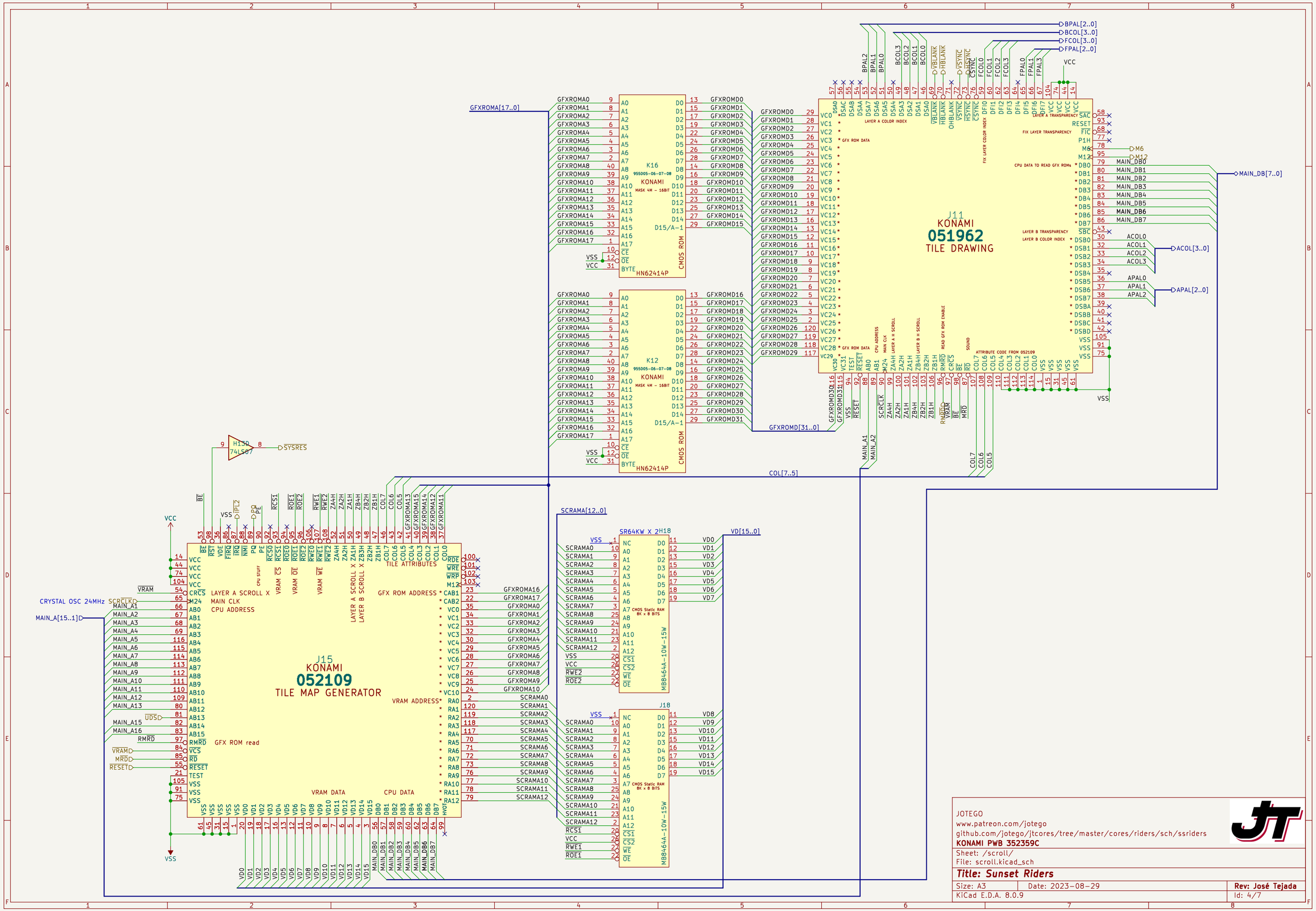
JOTEGO
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

Sheet: /colmix/
File: colmix.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 2/7


OBJROMA[18..0]
BK[7..0]
K1
74LS367
3-StateHexBuffers
L3
065A08
KONAMI
MASK 8M - 16BIT
CMOS ROM
054903
SHEET 18
MRW
OBJ
74LS32
D18A
NOT POPULATED
K3
F18D
74LS04
BK6 instead of VSS
when mounted (?)
J2
KONAMI
053245
OBJECT GENERATOR
OBJECT ROM ADDRESS BUS
MAIN ADDRESS BUS
OBJECT RAM ADDRESS BUS
OBJECT RAM DATA BUS
GENERATOR TO DRAWING
RAMOBJ
F19F
74LS04
OBJMPX
G2
G3
MB8464A-80-10-15/L-LL
CMOS Static RAM
8K x 8 BITS
RA[12..0]
RD[15..0]
NOT POPULATED
RA1 4kX8
RA2 4kX8
RA3 4kX8
RA4 4kX8
CN1
Conn_02x32_Row_Letter_First
DRUK0
J3
KONAMI
053244
OBJECT DRAWING
MAIN
PRIORITY CONTROL UNIT
E18A
74LS08
G15A
G15B
G15C
MAIN_DB[7..0]
G1
74LS245
buffers
G4
Quad Mux
74LS157
F4
HM6116P-3
2kB RAM
J1
MCM2018
SHEET 16
OCBNK0
OCBNK1
OCBNK2
MAIN_A[13..1]
MAIN_D[7..0]
OBJROMD[31..0]
AR[2..0]
BR[2..0]
HP[9..0]
PL[6..0]
SDIO
PR[14..13]
OCOL[8..0]
K7
JOTEGO
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C
Sheet: /objects/
File: objects.kicad_sch
Title: Sunset Riders
Size: A3
Date: 2023-08-29
KiCad E.D.A. 8.0.9
Rev: José Tejada
Id: 3/7

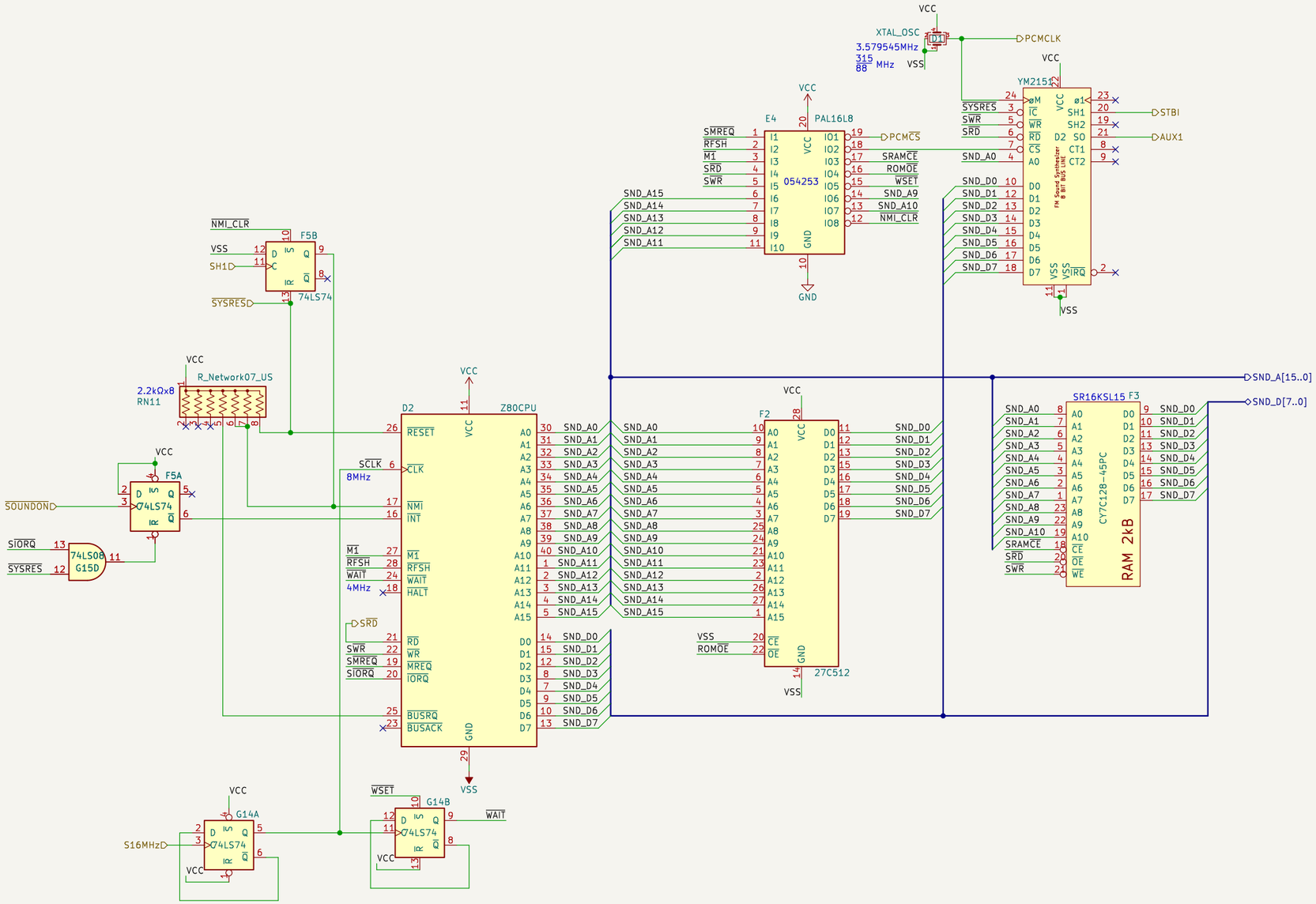


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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
KONAMI PWB 352359C

Sheet: /scroll/
File: scroll.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 4/7



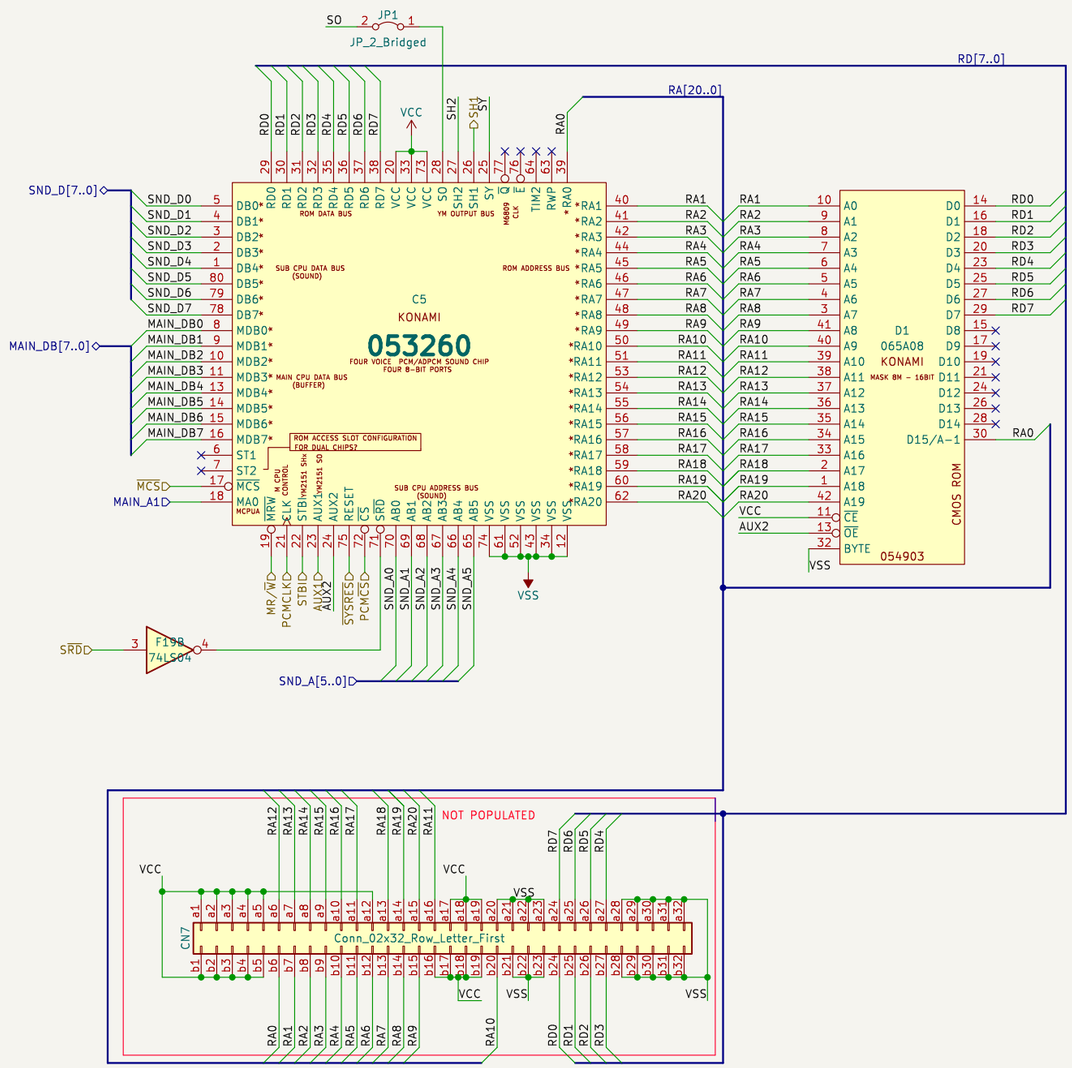
JOTEGO
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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders

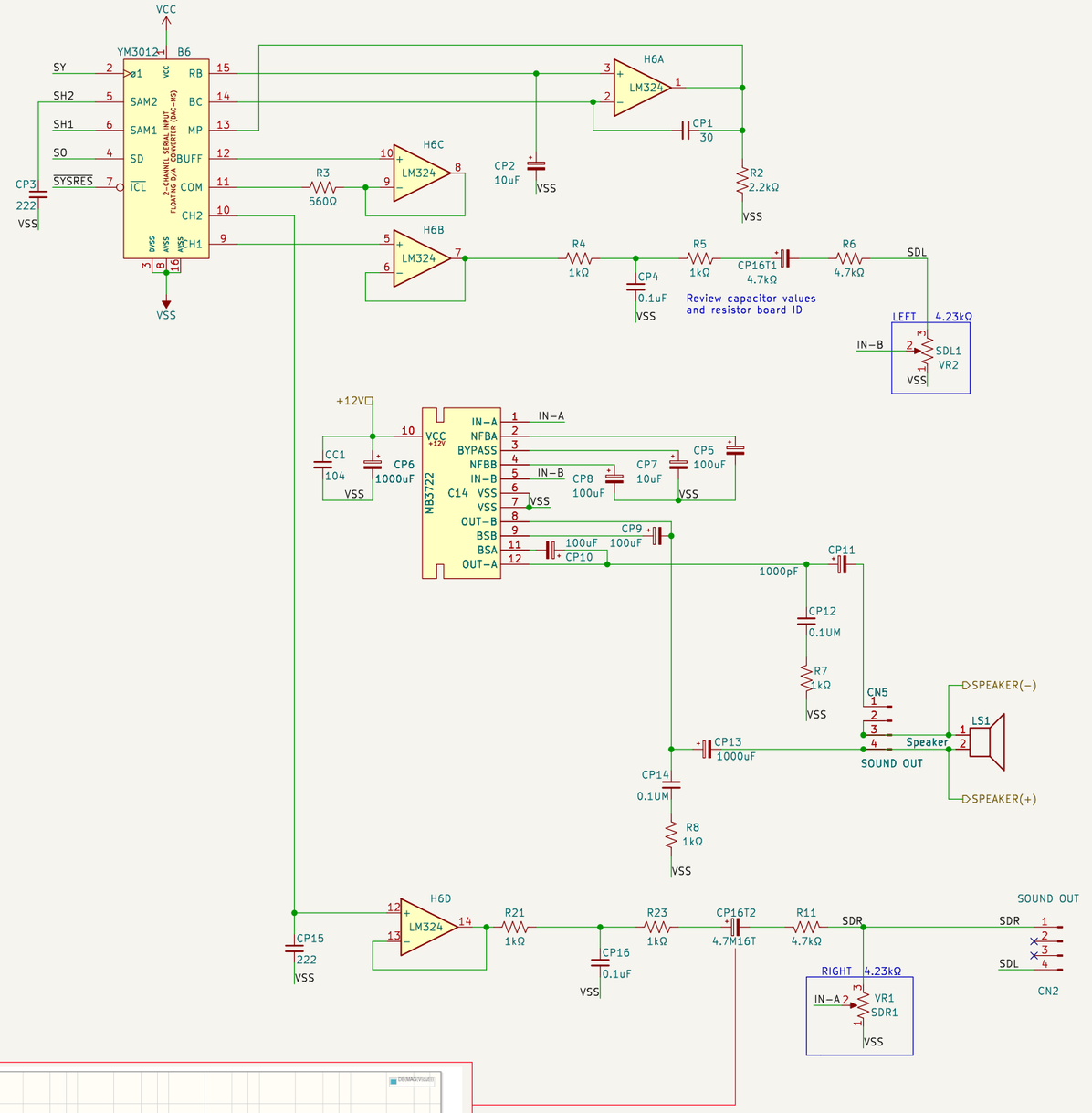
KONAMI PWB 352359C

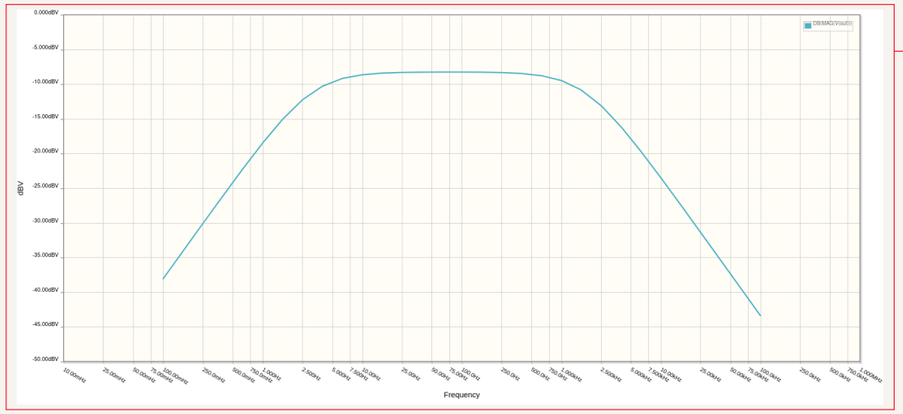
Sheet: /sound/
File: sound.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 5/7







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Sheet: /pcm_snd/		
File: pcm_snd.kicad_sch		
Title: Sunset Riders		
Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 6/7

main CPU data bus
to object RAM

D[15..0]
main CPU
data bus

RD[15..0]

K8
J8
74LS245
buffers
A->B
CE

H7
KONAMI
054358

-DMA access to main CPU 520us
-Returns a number based on
a CPU written RAM value
-updates obj priority data

19 pins top/bottom
13 pins left/right
64 pins total

MAIN_A[23..1]

to M68000

R1
1KΩ

F19F
74LS04
D17B
74LS00
E18C
74LS08
E18D
74LS08
D17A
74LS00
D17C
74LS00

OBJMPX
MRW
OBJSET
OBJ

PAL MAIN
SHEET 18

F5
PAL20RS10
054618

VPAD
OBJMPX
PDS
PQ

F15A
74LS74

MDTACK
VRAM
OBJ
OBJSET
COLOR
IO
MBOE
MRAMOEWR
MROMOE2
MROMOE1

SHEET 09

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github.com/jotego/jtcores/tree/master/cores/riders/sch/ssriders
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Sheet: /protection/
File: main2obj.kicad_sch

Title: Sunset Riders

Size: A3	Date: 2023-08-29	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 7/7